

Online/Data Supplement for

Wintertime Wood Smoke, Traffic Particle Pollution, and Preeclampsia

Vanessa Assibey-Mensah^{a,b,c}, J. Christopher Glantz^{a,d}, Philip K. Hopke^a, Todd A. Jusko^{a,e}, Kelly Thevenet-Morrison^a, David Chalupa^e, David Q. Rich^{a,e,f}

- a. Department of Public Health Sciences, University of Rochester Medical Center, Rochester, NY, USA
- b. Department of Obstetrics, Gynecology, and Reproductive Sciences, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA
- c. Magee-Womens Research Institute, University of Pittsburgh, Pittsburgh, PA, USA
- d. Department of Obstetrics and Gynecology, University of Rochester Medical Center, Rochester, NY, USA
- e. Department of Environmental Medicine, University of Rochester Medical Center, Rochester, NY, USA
- f. Department of Medicine, University of Rochester Medical Center, Rochester, NY, USA

Table S1. Distribution of air pollutant concentrations averaged in first gestational month during wintertime using land use regression, 2008-2013, Rochester, NY.

Pollutant	N	Mean	Standard Deviation	Minimum	5th	25th	50th	75th	95th	Maximum
BC (µg/m ³)	4,726	1.14	0.75	0.00	0.00	0.67	1.07	1.52	2.46	6.56
Delta-C (µg/m ³)	4,726	0.35	0.37	0.00	0.00	0.02	0.24	0.54	1.09	3.00
PM _{2.5} (µg/m ³)	4,361	6.9	3.5	0	2.4	4.7	6.3	8.4	13.0	47.8

Table S2. Pearson correlation coefficients for mean pollutant concentrations, relative humidity, and temperature for first gestational month during winter, 2008-2013 (n=4,350-4,726)

Pollutant/Weather Characteristic	BC	Delta-C	PM _{2.5}	Relative Humidity	Temperature
BC	1.00	--	--	--	--
Delta-C	0.26	1.00	--	--	--
PM _{2.5}	0.31	0.26	1.00	--	--
Relative Humidity	0.04	0.23	0.26	1.00	--
Temperature	0.03	-0.26	-0.32	-0.45	1.00

Table S3. Odds of preeclampsia associated with each interquartile range (IQR) increase in pollutant concentrations in each gestational month during the winter

Pollutant	Month	N	n of preeclampsia	OR (95% CI)
Black Carbon IQR=0.86 µg/m ³	1	4,652	220	1.16 (0.99, 1.36)
	2	4,992	223	1.12 (0.96, 1.32)
	3	5,203	229	1.18 (1.01, 1.38)
	4	5,177	225	1.06 (0.90, 1.25)
	5	5,165	238	1.25 (1.07, 1.45)
	6	4,822	238	1.08 (0.88, 1.33)
	7	4,494	223	1.11 (0.88, 1.39)
	8	4,133	215	1.07 (0.84, 1.35)
	9	3,617	167	0.86 (0.64, 1.16)
Delta-C IQR=0.52 µg/m ³	1	4,652	220	1.05 (0.87, 1.26)
	2	4,992	223	1.04 (0.88, 1.24)
	3	5,203	229	1.06 (0.89, 1.26)
	4	5,177	225	1.03 (0.87, 1.22)
	5	5,165	238	1.22 (1.02, 1.46)
	6	4,822	238	1.08 (0.89, 1.33)
	7	4,494	223	1.16 (0.93, 1.45)
	8	4,133	215	1.13 (0.89, 1.43)
	9	3,617	167	0.94 (0.71, 1.26)
PM_{2.5} IQR=3.64 µg/m ³	1	4,652	285	1.07 (0.92, 1.24)
	2	4,992	296	1.17 (1.01, 1.34)
	3	5,203	311	1.07 (0.94, 1.22)
	4	5,177	313	1.02 (0.88, 1.18)
	5	5,165	315	1.06 (0.92, 1.22)
	6	4,822	180	1.02 (0.88, 1.19)
	7	4,494	166	1.05 (0.90, 1.21)
	8	4,133	146	1.01 (0.86, 1.19)
	9	3,617	109	0.99 (0.81, 1.20)

N=sample size; n=cases

Adjusted for age, race/ethnicity, education, parity, multifetal gestation, year of conception, season of conception, pre-pregnancy diabetes, gestational diabetes, pre-pregnancy body mass index, birth hospital, relative humidity, and temperature

Table S4. Odds of early-onset preeclampsia associated with each interquartile range (IQR) increase in pollutant concentrations in each gestational month during the winter

Pollutant	Month	N	n of early-onset preeclampsia	OR (95% CI)
Black Carbon IQR=0.86 $\mu\text{g}/\text{m}^3$	1	4,418	51	1.54 (1.06, 2.25)
	2	4,752	56	1.38 (0.97, 1.96)
	3	4,951	59	1.40 (1.03, 1.92)
	4	4,926	62	1.04 (0.75, 1.46)
	5	4,911	61	1.17 (0.84, 1.65)
	6	4,682	40	1.11 (0.78, 1.56)
	7	4,356	28	1.13 (0.75, 1.70)
Delta-C IQR=0.52 $\mu\text{g}/\text{m}^3$	1	4,418	51	1.47 (1.01, 2.14)
	2	4,752	56	1.15 (0.81, 1.62)
	3	4,951	59	1.31 (0.97, 1.79)
	4	4,926	62	1.10 (0.81, 1.50)
	5	4,911	61	1.27 (0.93, 1.75)
	6	4,682	40	1.09 (0.77, 1.54)
	7	4,356	28	1.49 (1.04, 2.12)
PM_{2.5} IQR=3.64 $\mu\text{g}/\text{m}^3$	1	4,418	51	1.42 (1.11, 1.81)
	2	4,752	56	1.68 (1.36, 2.08)
	3	4,951	59	1.24 (1.06, 1.45)
	4	4,926	62	1.14 (1.00, 1.31)
	5	4,911	61	1.20 (1.02, 1.41)
	6	4,682	40	1.19 (1.01, 1.40)
	7	4,356	28	0.88 (0.61, 1.28)

N=sample size; n=cases

Adjusted for age, race/ethnicity, education, parity, multifetal gestation, year of conception, season of conception, pre-pregnancy diabetes, gestational diabetes, pre-pregnancy body mass index, birth hospital, relative humidity, and temperature

Table S5. Odds of late-onset preeclampsia associated with each interquartile range (IQR) increase in pollutant concentrations in each gestational month during the winter

Pollutant	Month	N	n of late-onset preeclampsia	OR (95% CI)
Black Carbon IQR=0.86 $\mu\text{g}/\text{m}^3$	1	4,601	234	1.04 (0.85, 1.26)
	2	4,936	240	1.10 (0.89, 1.36)
	3	5,144	252	1.13 (0.92, 1.38)
	4	5,115	251	0.87 (0.69, 1.08)
	5	5,104	254	1.10 (0.90, 1.34)
	6	4,782	140	1.08 (0.88, 1.33)
	7	4,466	138	1.08 (0.87, 1.34)
	8	4,113	126	1.09 (0.88, 1.35)
	9	3,616	108	1.00 (0.78, 1.28)
Delta-C IQR=0.52 $\mu\text{g}/\text{m}^3$	1	4,601	234	1.00 (0.82, 1.22)
	2	4,936	240	1.13 (0.93, 1.37)
	3	5,144	252	1.04 (0.85, 1.27)
	4	5,115	251	1.00 (0.81, 1.23)
	5	5,104	254	1.13 (0.92, 1.38)
	6	4,782	140	1.20 (0.98, 1.46)
	7	4,466	138	1.23 (0.99, 1.51)
	8	4,113	126	1.20 (0.98, 1.48)
	9	3,616	108	1.18 (0.93, 1.50)
PM_{2.5} IQR=3.64 $\mu\text{g}/\text{m}^3$	1	4,601	234	1.00 (0.85, 1.19)
	2	4,936	240	1.03 (0.87, 1.23)
	3	5,144	252	0.98 (0.83, 1.16)
	4	5,115	251	0.96 (0.80, 1.14)
	5	5,104	254	1.02 (0.86, 1.20)
	6	4,782	140	0.96 (0.80, 1.15)
	7	4,466	138	1.07 (0.93, 1.24)
	8	4,113	126	1.04 (0.88, 1.22)
	9	3,616	108	1.00 (0.82, 1.22)

N=sample size; n=cases

Adjusted for age, race/ethnicity, education, parity, multifetal gestation, year of conception, season of conception, pre-pregnancy diabetes, gestational diabetes, pre-pregnancy body mass index, birth hospital, relative humidity, and temperature